

Practice — The Magnetic Effect of Electric Current

Section A is interactive — tap an option or type and press Check. Sections B and C are written practice.

Objective (1 mark each)

Q1 • MCQ**[1]**

When current flows through a wire, a nearby compass needle —

- (a) melts (b) deflects (c) disappears (d) does nothing

Q2 • MCQ**[1]**

The link between electricity and magnetism was discovered by —

- (a) Volta (b) Galvani (c) Oersted (d) Hooke

Q3 • MCQ**[1]**

The region around a wire where its magnetic effect is felt is called the —

- (a) electric circuit (b) magnetic field (c) electrolyte (d) conductor

Q4 • TRUE / FALSE**[1]**

The magnetic field around a wire remains even after the current is switched off.

- (a) True (b) False

Q5 • ASSERTION-REASON**[1]**

A: A compass needle deflects near a current-carrying wire.

R: A current produces a magnetic field around the wire.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Q6 · FILL IN THE BLANK

[1]

Producing a magnetic field by passing current through a conductor is called the _____ effect of electric current.

Q7 · FILL IN THE BLANK

[1]

Oersted made his discovery in the year _____.

Short answer (2 marks each)

Q8

[2]

State what is observed in Activity 4.1 when the switch is turned ON and then OFF.

Q9

[2]

Define magnetic field and name one device that uses the magnetic effect of current.

Q10

[2]

What did Oersted discover, and how?

Long / application (3 marks each)

Q11

[3]

Describe an experiment (with a cell, switch, wire and compass) to show that an electric current has a magnetic effect. State your observation and conclusion.

Q12

[3]

A compass below a current-carrying wire deflects to one side. Predict and explain what happens if the battery terminals are reversed.

[Check yourself](#) → Answer key